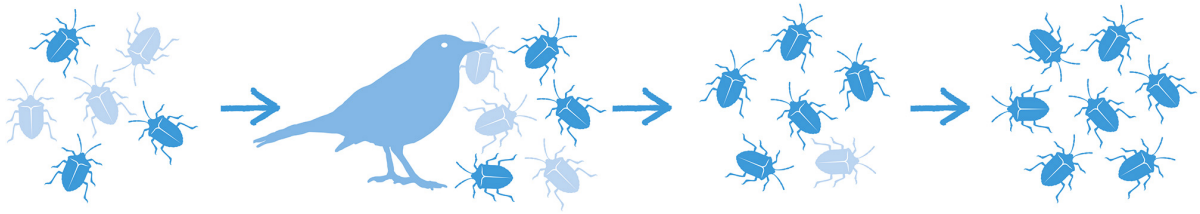
Natural selection



There is variation in traits. For example, some beetles are pale and others dark.

There is differential reproduction. No environment can support unlimited population growth, so some individuals lose out. Here, birds eat the pale beetles, so fewer of them reproduce.

There is heredity. The dark beetles have more dark offspring because this trait has a genetic basis.

End result: If darkness is the winning trait, producing more offspring, in time, all beetles will be dark.

Read accessible description

“I see no good reasons why the views given in this volume should shock the religious views of anyone.”

Charles Darwin

Sexual selection

Darwin also developed a theory of sexual selection. First outlined in *On the Origin of Species*, this was developed further in *The Descent of Man, and Selection in Relation to Sex* (1871). This theory was distinct from natural selection, as Darwin recognized that animals select mates based on characteristics that do not simply favor survival. For example, when Darwin considered the spectacular but cumbersome tails of male peafowl (peacocks), he could not imagine the tail playing any role in helping the individual bird to survive. He concluded that they were designed to boost an individual’s chance of reproductive success. Peahens choose males with the brightest tails, so the genetic material of these showy males is passed to the next generation. Bright tail feathers indicate that the bird is healthy, so choosing a mate with a bright tail is a good strategy for the peahen. However, Darwin’s idea that females choose a mate came under fire; 19th-century European society could accept that males competed to reproduce